CAREER GLITCH · SHORT · SIMPLE · NO BORING PARTS

ACTUARY IN THE AI ERA

A model can calculate. An actuary still chooses the assumptions, understands the regulation and signs off on the risk.

★ TL;DR — Median ~$125,000 with a 2–4 year study band through mathematics, professional examinations and credentialing. The catch: the exam marathon begins before the first job and continues while you work. ★

THE CAREER GLITCH TAKE

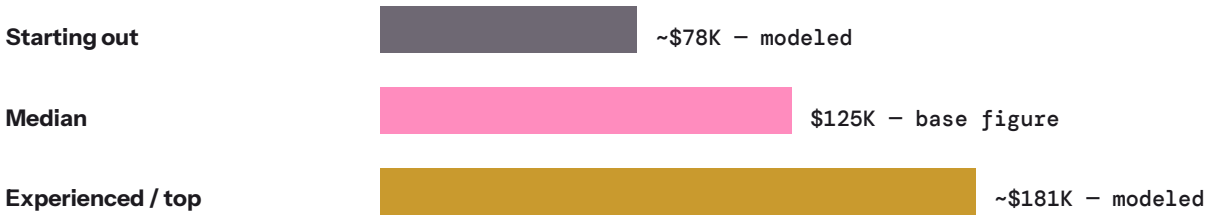
OUR TAKE: YES

MEDIAN ~$125K · TRAINING 2-4 years · AI RISK MEDIUM · STATE LICENSE

- The credential is difficult, which is exactly why it protects the pay.
- AI speeds data work, but assumptions and accountability remain professional judgment.
- This is a marathon for people who genuinely like probability and uncertainty.

The site's take, built from this profession's numbers in the Career Glitch base.

THE MONEY — LOOK AT THE BARS



The Career Glitch base figure for Actuary is a ~$125K median (compiled from U.S. Bureau of Labor Statistics, Occupational Outlook Handbook ranges). The starting and top bars are modeled around that median — orientation, not a promise. Pay swings hard by state, employer and experience: check live postings in your state before planning around any number.

WHAT DOES A ACTUARY ACTUALLY DO?

✓ Analyze data about mortality, illness, accidents, property loss or investments

- ✓ Build models that estimate future costs and financial risk
- ✓ Price insurance products, set reserves and test business scenarios
- ✓ Explain assumptions and results to managers, regulators and clients

Think: a professional odds-maker for an uncertain future, except the bets involve pensions, insurance and real financial promises.

WHAT AI ALREADY TOOK (AND WHAT IT CAN'T)

THE ROBOT	THE HUMAN
AI can clean data, generate model code, summarize results and automate routine reporting.	Actuaries still choose assumptions, judge whether data fits the question, satisfy professional standards and explain uncertainty without hiding behind a model.

⚠ **Do not choose this only for the salary. The professional exams are a long filter, and weak interest in mathematics will make every stage feel like punishment.**

THE VIBE — IS THIS YOUR LIFE?

THE JOB IS	THE JOB IS NOT
✓ Remote-capable analytical work with strong professional status ✓ Clear credential milestones and a respected specialty ✓ Problems that connect mathematics with real business decisions	✗ Years of studying for exams alongside full-time work ✗ Regulatory and documentation demands are substantial ✗ Progress can feel slow when an exam attempt does not pass

Good fit? You love probability, statistics and careful reasoning, and you can keep studying after formal school ends.
Bad fit? You dislike mathematics, need fast credential payoff or want work where assumptions do not matter.

THE FORK — PICK ONE DOOR

DOOR A — MATHEMATICS DEGREE PLUS EXAMS	DOOR B — INSURANCE ANALYTICS BRIDGE
Build a degree-level foundation in mathematics, statistics, economics and programming, then pass initial professional exams and compete for actuarial internships or entry roles.	Start in insurance analytics, underwriting support, finance or data analysis while passing exams. The bridge is credible when the work uses real risk data and your exam progress is visible.

Smart move: Pass an initial professional exam and build an Excel, SQL and Python portfolio before applying. That combination answers the employer's two fears: can you pass, and can you work?
Where you land after: Credentialed actuary, then actuarial manager, consultant, product or risk leader.

YOUR ROADMAP — 6 STEPS

1	2	3	4	5	6
BUILD THE	START THE	CREATE APPLIED	ENTER THROUGH	CONTINUE CREDENTIALI	SPECIALIZE AND

STEP 1 — Build the mathematics base

STUDY · FOUNDATION

- Master probability, statistics, calculus, economics and financial mathematics.
- Learn Excel deeply, then add SQL and Python or R for data work.

STEP 2 — Start the professional exams

REQUIRED · CREDENTIAL GATE

- Choose the actuarial pathway that matches property, life, health or pension work.
- Prepare seriously for the first examinations and treat a pass as career evidence.

STEP 3 — Create applied proof

PORTFOLIO · REAL DATA

- Build a pricing, reserving or risk-analysis project with documented assumptions.
- Show sensitivity tests and a plain-language explanation of what the result means.

STEP 4 — Enter through internship or analytics

PAID · FIRST ROLE

- Apply to actuarial internships and entry roles with exam progress on your résumé.
- Use adjacent insurance analytics work when it keeps you close to actuarial teams.

STEP 5 — Continue credentialing at work

LONG GAME · WHILE EMPLOYED

- Progress through the associate-level requirements for your chosen actuarial society.
- Rotate through pricing, reserving, modeling or valuation to broaden judgment.

STEP 6 — Specialize and lead

EARNED · ADVANCED LEVEL

- Complete fellowship or the advanced credential path appropriate to your specialty.
- Move toward product, consulting, risk leadership or management as your interests dictate.

FIRST 90 DAYS — JUST DO THIS

□ Today: choose the actuarial specialty that interests you and map its examination pathway.

□ Days 1–30: begin probability study and build spreadsheet and programming fundamentals.

□ Days 31–60: work through a full exam syllabus section and complete a small risk-modeling project.

□ Days 61–90: schedule or prepare to schedule your first professional exam and apply for relevant internships or analytics roles.

□ Day 90: you should have measurable exam progress and applied work to show, not only an intention to become an actuary.

NEXT — MAKE IT PERSONAL

This roadmap is the same for everyone. Your life is not.

- Not sure this career fits YOU? Take the Big Career Test — 50 questions, your top-3 matches and one honest "probably NOT you". → careerglitch.com/#test
- Already sure? Turn AI into Your Personal Career Mentor — a 2,300-word system that interviews you first, checks your hours, budget and deadline honestly, and builds YOUR plan into a Notion dashboard. → careerglitch.com/#prompt

★ **PRICE THE RISK HONESTLY.** ★